

Supplementary Material: Challenges of Gate-Based Quantum Optimization for a Discrete Process-Design Problem

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ABSTRACT

This supplement records provenance, validation, and artifact anchors for the main manuscript on gate-based quantum optimization of a drug-substance manufacturing (DS-MFG) discrete process-design problem. It is intended to be compiled and submitted separately from the main conference paper if appendices are not counted as article body material in the submission workflow.

Keywords: supplementary material, quantum optimization, process design, reproducibility

REPOSITORY AND PACKAGING

The main manuscript keeps only the narrative-critical values. This supplement keeps the audit data close to the manuscript source while allowing the main article to remain within the conference page limit. The repository URL is <https://github.com/SECQUOIA/pse-quantum-discrete-optimization>. A final archival DOI has not yet been assigned. It will be added before camera-ready submission if one is minted; the review artifact is identified by the repository snapshot that contains this supplement and by the checksums below.

Table 1: Current reproducibility pins validated for the DS-MFG case study.

Layer	Current pin or check	Repository anchor
Julia optimization wrapper	QiskitOpt.jl v0.7.0; manifest git tree a601dcccdd43fd0a8b18e63f3b771273817af49eb.	case-studies/ds-mfg-qubo-qiskitopt/Project.toml; case-studies/ds-mfg-qubo-qiskitopt/Manifest.toml.
Julia QUBO stack	QUBODrivers.jl v0.6.1, QUBOTools.jl v0.13.0, QUBO.jl v0.6.0.	case-studies/ds-mfg-qubo-qiskitopt/Project.toml; case-studies/ds-mfg-qubo-qiskitopt/Manifest.toml.
Python bridge	Qiskit 2.3.1, Qiskit Aer 0.17.2, Qiskit Optimization 0.7.0, Qiskit IBM Runtime 0.46.1, NumPy 2.2.6, SciPy 1.15.3.	case-studies/ds-mfg-qubo-qiskitopt/CondaPkg.toml.
Fresh-clone validation	julia --project=. scripts/smoke_test.jl imports packages, checks required artifacts, and validates cached summary values.	case-studies/ds-mfg-qubo-qiskitopt/scripts/smoke_test.jl.

Table 2: Reproducibility levels used for the DS-MFG manuscript package.

Level	Meaning	Current support
Score-from-cache reproducible	Retained samples, summaries, and repair tables can be inspected and rescored without rerunning the stochastic campaigns.	Supported by cached distributions, hit-rate reports, checksums, and the smoke test. This is the main reproducibility level for the manuscript claims.
Rerunnable with retained scripts	The scripts and pinned environment can rerun selected QAOA/VQE, classical, and hardware-handoff workflows, subject to stochastic variation and credential availability.	Partially supported. QAOA fixed-angle parameters and sampled outputs are retained; selected VQE metadata reruns now retain optimized ansatz vectors for seed 74018 and for the final-budget selected seeds 74018, 74007, and 74001.
Full campaign replay	Every stochastic campaign can be reproduced byte-for-byte from retained optimizer states, simulator logs, hardware calibration, and submitted-circuit metadata.	Not claimed. Some optimizer, MPS truncation, and submitted-job hardware metadata were not retained.

Table 3: Checksums for retained archives and primary cached summaries.

Artifact	SHA-256 checksum
case-studies/ds-mfg-qubo-qiskitopt/Fw_DSfmfgcasequboinformation.zip	aa20d793fc45c25c2bd7cc0a231877f71b225bf80daee1153lcef6eb6cac5af2
ds_mfg_ibm_qaoa_pilot_fez_4096x3.zip	a3f5441d81a107ff99e7df3a1ab4b0e1ef4df6de813ed3416d4997d1d6262b83
ds_mfg_direct_full_qubo_hardware_pilot.zip	25a39b061f588523eeb609c397abc3b2afd8dd4f15c4f9d6d8037772532c6515
case-studies/ds-mfg-qubo-qiskitopt/ds_mfg_qaoa_juliquaoa_transfer_highread/qaoa_juliquaoa_transfer_summary.csv	10bb3bc9b746fb12a87eadc1fd2b819c1a3cacbc84025c9bd54797c1088c34b7
case-studies/ds-mfg-qubo-qiskitopt/ds_mfg_vqe_reduce_d_flow_objective_final/vqe_reduced_top50_sampling_summary.csv	aac0299d49314b8d0e4d54b088f50cc728f1d212540613348ab62d209a132ee1
case-studies/ds-mfg-qubo-qiskitopt/ds_mfg_gurobi_provenance/ip_exact_enumeration_summary.json	1298dcb07aa82c0adc05ead3fb1b3f4b86ae9ef38e9a276b98f97ec5a7606e5e
case-studies/ds-mfg-qubo-qiskitopt/ds_mfg_gurobi_provenance/gurobi_local_pool_rerun_summary.json	13f7683c41cbf14b4225326e41778c6f42ae3979e4d1303af65c7132fe35a6e7
case-studies/ds-mfg-qubo-qiskitopt/ds_mfg_vqe_reduce_d_flow_objective_selected_metadata/vqe_reduced_top50_sampling_summary.csv	df82de49b921e3b70d5d24f0896e77b6da64477c3a13cffc0a0761d6e9ebd2db
case-studies/ds-mfg-qubo-qiskitopt/ds_mfg_hit_rate_reports/time_to_solution_report.csv	c27e70684f3b753efc4eb8cf2b75e79afe4b15402bcd6b3ca7d7a598310f2efd

VALIDATION FACTS

Table 4: Numerical validation facts used to support the manuscript claims.

Check	Retained value	Anchor
Exact repaired optimum	$F(y^*) = 11.7095$ at flow 1001110100111100011.	ds_mfg_reduced_flow_objective/reduced_flow_summary.csv; ds_mfg_reduced_flow_objective/reduced_exact_top_flows.csv.
Gurobi pool bridge	All 36 stored Gurobi-pool flows match the repaired QUBO score within 3.1×10^{-13} .	ds_mfg_reduced_flow_objective/reduced_exact_top_flows.csv; ds_mfg_gurobi_provenance/gurobi_local_pool_rerun_solutions.csv.
Retained Gurobi metadata	Model capex: 19 binary variables, 20 constraints, 58 nonzeros; PoolSearchMode=2, PoolSolutions=36, solve time 0.073 s, 36 solutions found. Solver version, status code, MIPGap, PoolGap, and pool completeness/diversity intent were not retained in the original export. A local Gurobi 13.0.2 rerun with PoolSearchMode=2 and PoolSolutions=100 returned OPTIMAL, MIPGap=0.0, 36 solutions, and exact retained-pool agreement.	Fw_DSfmfgcasequboinformation.zip/gurobi_pool_solutions_n36_20260126_105706.json.
Quadratic surrogate weakness	Global fit $R^2 \approx 0.999$, but the surrogate optimum repairs to 14.5505. The true repaired optimum is surrogate rank 33; the surrogate optimum is exact repaired rank 6.	ds_mfg_reduced_flow_objective/reduced_flow_summary.csv; Figure 1.
Low-energy surrogate audit	Exact enumeration finds 36 feasible flows (the complete pool). The 50 lowest-cost repaired flows include 14 infeasible flows — the penalty weights are insufficient to push all infeasible flows above all feasible ones. In the lowest 1% repaired-flow set, exact-rank versus surrogate-rank Spearman correlation is 0.901. Exact enumeration wall time was not retained.	ds_mfg_reduced_flow_objective/reduced_exact_top_flows.csv; MANUSCRIPT_FINDINGS.tex.
Primary QAOA simulator check	JuliQAOA statevector probability on the reference optimum was 2.63×10^{-3} for the energy-targeted $p = 5$ angles. The cached Aer MPS sampled rate was 2.53×10^{-3} , inside its Wilson 95% interval $[2.35, 2.73] \times 10^{-3}$. MPS truncation logs were not retained.	ds_mfg_qaoa_juliqaoa_angle_search/juliqaoa_angle_summary.csv; ds_mfg_qaoa_juliqaoa_transfer_highread/qaoa_juliqaoa_transfer_summary.csv.

Table 5: Implications of missing provenance values for the manuscript claims.

Missing value	Affects main claim?	Affects full rerun?	Current mitigation
Historical Gurobi final status and MIPGap	No for the current claim after rerun; the original export still lacks these fields, but the model/pool behavior has been rerun locally.	Partly; exact historical solver state is unavailable, but current solver provenance is retained.	The same flow is certified by exact enumeration over all 2^{19} flows and by a local Gurobi 13.0.2 rerun with <code>OPTIMAL</code> status, <code>MIPGap=0.0</code> , 36 solutions, and exact retained-pool agreement.
MPS truncation logs	No for the central QAOA sampling claim.	Yes, for simulator-forensics replay.	The energy-targeted QAOA row is cross-checked against independent JuliQAOA statevector probabilities, which agree with the Aer MPS sampled rate within the Wilson interval.
VQE optimized parameter vectors	No; VQE is secondary and reported as a retained sampled distribution.	Partly; selected fixed-circuit replay is now supported, but full historical optimizer replay is not claimed.	Metadata reruns retain optimized <code>EfficientSU2</code> vectors for seed 74018 and for seeds 74018, 74007, and 74001 at the final sampling budget. The current metadata rerun produced 4 reference hits total and is provenance, not a replacement for the historical 28-hit final follow-up.
Submitted hardware layout, depth, and two-qubit count	No, because the hardware row is a submission-and-scoring handoff only.	Yes, for hardware-performance diagnosis.	The manifest retains backend, job IDs, shots, seeds, logical circuit metadata, and scored counts. No noise-mechanism claim is made.
Hardware calibration or mitigation metadata	No under the current claim.	Yes, for calibrated hardware analysis.	The manuscript states that no read-out or error mitigation is claimed and that the retained metadata cannot diagnose a specific noise source.
Exact enumeration wall time	No.	Partly, for cost accounting.	The enumeration is documented as exact over all 2^{19} flows; issue 24 tracks recording wall time if the enumeration script is rerun.

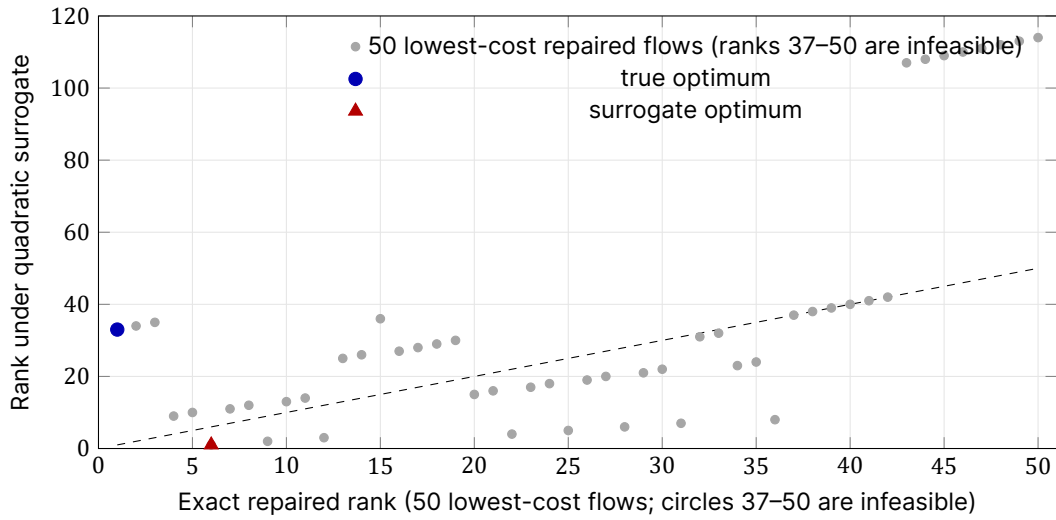


Figure 1. Surrogate rank audit for the 50 lowest-cost repaired flows. Exact enumeration confirms there are exactly 36 feasible flows (ranks 1–36); the remaining 14 shown here are infeasible flows whose penalty values are insufficient to place them above all feasible ones. The surrogate has high global R^2 but imperfect low-energy ordering: the exact optimum is only rank 33 under the surrogate, while the surrogate minimum repairs to the sixth-best exact flow. Quantum-sampling outcomes are therefore scored using the exact repaired objective F , not the surrogate $\hat{F}$.

EXACT AUXILIARY REPAIR: COMPONENT DECOMPOSITION

The QUBO objective $E_{\text{QUBO}}(y, z)$ acts on 36 binary variables: the 19 flow (design) variables $y = (y_1, \dots, y_{19})$ and 17 binary auxiliary (slack) variables $z = (z_1, \dots, z_{17})$ introduced by the penalty reformulation. A sampled 36-bit string is not directly interpretable as a flowsheet because the auxiliary bits are bookkeeping for the inequality constraints of the original design problem, not design decisions. After projecting a sample onto the 19 flow bits y , the repair step finds the auxiliary assignment that minimizes the QUBO energy for that flow:

$$z^*(y) = \arg \min_{z \in \{0,1\}^{17}} E_{\text{QUBO}}(y, z). \quad (1)$$

Brute-force enumeration over all $2^{17} = 131,072$ auxiliary assignments is tractable for a single call but unnecessarily costly when repair is applied to every sampled state across all workflows. Once y is fixed, every term in E_{QUBO} that involves an auxiliary variable z_k is either (i) a linear term $c_k z_k$, (ii) an auxiliary-auxiliary coupling $c_{k\ell} z_k z_\ell$, or (iii) a flow-auxiliary coupling $c_{ik} y_i z_k$ (a constant given y). Consider the *auxiliary coupling graph*: nodes are the 17 auxiliary variables, and an edge connects z_k and z_ℓ whenever their quadratic coefficient $c_{k\ell} \neq 0$. For this QUBO the graph has 13 connected components: 9 isolated nodes (one auxiliary bit each) and 4 edges (two auxiliary bits each). Because there are no edges between components, fixing y makes the components conditionally independent, and (1) decomposes as

$$\min_z E_{\text{QUBO}}(y, z) = \text{const}(y) + \sum_{c=1}^{13} \min_{z^{(c)}} E^{(c)}(y, z^{(c)}), \quad (2)$$

where $z^{(c)}$ are the auxiliary bits in component c and $E^{(c)}$ contains only the terms that involve those bits (with fixed y values substituted as constants). Each sub-problem is solved by enumerating $2^1 = 2$ or $2^2 = 4$ assignments and selecting the minimum energy; the total work is at most $9 \times 2 + 4 \times 4 = 34$ comparisons per repair, a factor of $\sim 3,800\times$ smaller than the brute-force bound. This procedure is implemented in `scripts/run_classical_baselines.jl`; the component structure is read from `ds_mfg_reduced_flow_objective/auxiliary_components.csv`.

Physical meaning of the components. Table 6 lists all 13 components with their auxiliary variable indices, the flow variables whose fixed values enter each local energy, and a brief description. Singleton components (1–5, 10–13) correspond to individual inequality constraints in the penalty QUBO: the single binary slack takes the value (0 or 1) that satisfies the inequality, which for feasible

flow assignments is the constraint residual. Pair components (6–9) each arise from the product-linearization of the Route B activation rule ($f_{06} = 1$ iff Reactor 1 is PFR or CSTR and Reactor 2 is Batch). Linearizing a product of two binary variables requires up to three inequality constraints; the slack for such a constraint can take values in $\{0, 1, 2\}$, needing two binary bits. The four pair components together represent the complete linearization of this one coupling rule across all Reactor 1/Reactor 2 type combinations.

Table 6: Independent auxiliary components for the DS-MFG QUBO repair (source: `ds_mfg_reduced_flow_objective/auxiliary_components.csv`). Variable indices 1–19 are the 19 flow variables y ; indices 20–36 are the 17 binary auxiliary (slack) variables z . *Flow neighbors* are the y -indices that appear in the local energy $E^{(c)}$. No coupling term links auxiliaries across components, so each sub-problem is solved independently by enumeration.

#	Aux. idx.	n_{aux}	Flow neighbors (y idx. and f -arc)	$2^{n_{\text{aux}}}$	Role
1	20	1	17 (f_{16} , Batch U/S), 18 (f_{17} , cont. exit)	2	Batch/cont. downstream exclusion
2	21	1	2 (f_{01} , PFR Rxn 1), 3 (f_{02} , CSTR Rxn 1), 4 (f_{03} , Batch Rxn 1)	2	Reactor 1 at-most-one
3	22	1	6 (f_{05} , Route A), 7 (f_{06} , Route B)	2	Route at-most-one
4	23	1	9 (f_{08} , PFR Rxn 2), 10 (f_{09} , CSTR Rxn 2), 11 (f_{10} , Batch Rxn 2)	2	Reactor 2 at-most-one
5	24	1	14 (f_{13} , Cont. I), 15 (f_{14} , Cont. II), 16 (f_{15} , Cont. III), 17 (f_{16} , Batch U/S)	2	Downstream at-most-one
6	25, 26	2	2 (f_{01} , PFR Rxn 1), 7 (f_{06} , Route B), 11 (f_{10} , Batch Rxn 2)	4	Route B linearization (PFR Rxn 1 case)
7	27, 28	2	3 (f_{02} , CSTR Rxn 1), 7 (f_{06} , Route B), 11 (f_{10} , Batch Rxn 2)	4	Route B linearization (CSTR Rxn 1 case)
8	29, 30	2	4 (f_{03} , Batch Rxn 1), 7 (f_{06} , Route B), 11 (f_{10} , Batch Rxn 2)	4	Route B linearization (Batch Rxn 1 case)
9	31, 32	2	2 (f_{01} , PFR Rxn 1), 3 (f_{02} , CSTR Rxn 1), 6 (f_{05} , Route A)	4	Route B linearization (Route A side)
10	33	1	6 (f_{05} , Route A), 11 (f_{10} , Batch Rxn 2)	2	Route A / Rxn 2 coupling
11	34	1	14 (f_{13} , Cont. I), 18 (f_{17} , cont. exit)	2	Cont. I flow balance
12	35	1	15 (f_{14} , Cont. II), 18 (f_{17} , cont. exit)	2	Cont. II flow balance
13	36	1	16 (f_{15} , Cont. III), 18 (f_{17} , cont. exit)	2	Cont. III flow balance
9×2 + 4×4 = 34 comparisons per repair vs. $2^{17} = 131,072$ brute force (~3,800× reduction).					

VARIATIONAL AND HARDWARE PROVE- NANCE

Table 7: Variational-parameter provenance for the central QAOA and VQE rows.

Workflow	Circuit / ansatz	Objective and optimizer	Parameter source	Validation metric
Energy-targeted QAOA	$p = 5$ QAOA on the 19-flow surrogate.	Minimize $\mathbb{E}_{p_\theta}[\hat{F}(y)]$ with JuliQAOA find_angles_bh; seed 91001; 5 basin-hopping iterations.	ds_mfg_qaoa_juliqaoa_angle_search/juliqaoa_angle_summary.csv; angles transferred beta-then-gamma to QiskitOpt.QAOA.	664 reference hits / 262144 reads; 39661 top-50 reads.
Top-10 QAOA diagnostic	$p = 5$ QAOA on the 19-flow surrogate.	Maximize probability on the enumerated exact top-10 set; oracle-aided because the repaired ranking is known.	Top-10 angle vector retained in ds_mfg_ibm_qaoa_pilot_fez_4096x3.zip/job_manifest.json; local 1007-hit diagnostic CSV was not retained.	1007 reference hits / 262144 reads in the retained manuscript summary; hardware handoff used this circuit.
Reduced-surrogate VQE	EfficientSU2; 152 initial parameters on the 19-flow surrogate.	COBYLA through QiskitOpt.VQE; 128 optimizer reads; 25 iterations; seeds 74001, 74007, and 74018 in the final follow-up.	Historical sampled distributions retained; current metadata reruns persist optimized ansatz vectors in ds_mfg_vqe_reduced_flow_objective_seed_74018_metadata/ and ds_mfg_vqe_reduced_flow_objective_selected_metadata/.	Historical final follow-up: seed 74018 had 20 reference hits / 524288 reads; three seeds had 28 reference hits / 1572864 reads. Current metadata rerun: 4 reference hits / 1572864 reads.

Table 8: Hardware pilot provenance retained in the submitted-job archive.

Item	Retained metadata	Limitation
Backend and jobs	ibm_fez; backend queried 2026-06-17 20:19 UTC; 156 qubits; 9 completed jobs; 4096 shots per job; repeats 1–3; transpile seeds 92001–92003; job IDs retained.	Queue time, calibration snapshot, readout mitigation, and error mitigation metadata were not retained.
Logical circuit	19 qubits and 19 classical bits; pre-transpile depth 86; operation counts: 19 h , 95 rx , 95 rz , 255 rz , and 19 measurements; scoring reverses Qiskit count keys to $x_1, \dots, x_{19}$.	Submitted-job transpiled depth, two-qubit-gate count, physical layout, and measurement map were not retained.
Hardware outcome	36864 total reads; 6 top-50 reads; 1 top-10 read; 4 stored Gurobi-pool reads; 0 reference-optimum reads; best repaired flow rank 2 with objective 11.8105. With zero reference hits, the two-sided 95% Wilson upper bound on the reference-optimum rate is 1.04×10^{-4} .	The pilot validates submission and scoring only; it cannot diagnose a specific noise mechanism.